data that can be useful in finding lost or crashed aircraft.

Fig. 2 is an example of active remote sensing. It may appear as just a light-and-dark pixel image. But after analysing the same, a lot of information can be found. The dark spot in the image is due to specular reflection. In this case, it shows smooth surfaces like paved surfaces or a flowing river. Double bounce backscatter waves in the image represent the bright white part in the centre. That is an urban feature like a building. Majority of the radar image is due to diffused scattering, that is, from the rough surface of vegetation growing in agricultural areas.

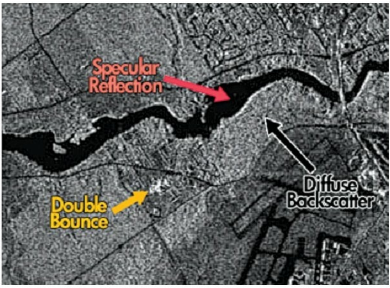

Fig. 2: Image captured using active remote sensing through Radarsat-2 (Credit: https://gisgeography.com)

Reflected light in passive remote sensing provides images in the whole electromagnetic spectrum. For example, the multispectral image shown in Fig. 3 can have different band combinations; the bright white shows built-up areas, while the darkest shade shows water.

Integrated sensors in weather monitoring systems

India has launched many weather-monitoring satellites to monitor the weather and climate of the Earth. INSAT-3DR, the latest weather-monitoring satellite, carries Satellite Aided Search and Rescue Transponder. It picks up and relays alert signals originating from distress beacons on land, air or sea. It provides better night-time pictures of clouds and fog, and sea surface temperature can be measured with better accuracy. The satellite also estimates precipitation, cloud motion, winds, humidity, temperature, snow cover and so on. Images help capture atmospheric haze, different land cover types such as forests, croplands, deserts, inland water bodies, glaciers, grasslands, wetlands and the like in optical and thermal bands.

INSAT Meteorological Data (IMD) Data Processing System (IMDPS), operational at IMD, New Delhi, disseminates the data from meteorological satellites. All over the country many automatic weather stations (AWSes) have been installed by IMD and other agencies. IMD has also installed Automatic Rain Gauge (ARG) stations. An AWS requires different types of sensors to monitor barometric pressure, solar radiation, wind speed, wind direction, humidity, temperature, pH, conductivity, dissolved oxygen and more.


Fig. 3: Image captured using passive remote sensing through Radarsat-2 (Credit: https://gisgeography.com)

Military and satellites both utilise sunlight as a power source using solar panels. This is also a renewable source of energy. Nikhil Verma, engineer at BKC Weather System, quotes, "Kipp and Zonen has recently launched a new sensor, that is, DUST-IQ optical sensor, which helps calculate the efficiency of the solar panels while setting up a solar power plant (SPP). Analysis of the data of environmental parameters like radiations, wind speed, wind direction, temperature, relative humidity, barometric pressure and rainfall helps calculate the performance ratio of an SPP. This also helps achieve better performance and enhance the power generation of an SPP."

Sanjay More, manager at Skymet Weather Services, says that, a tipping bucket rain gauge is the latest sensor that helps measure rainfall, as it is a crucial parameter to monitor for most applications. The latest technology is to use efficient algorithms at an AWS to validate the values or data of any measurement method, to check if it is true or false.

Topography is the main challenge in weather monitoring through satellites, so we need to establish a proper AWS to compare data with satellite information. To establish proper networking between an AWS and satellite stations, accurate information is needed. Skymet is India's largest weather monitoring and forecast solution provider.

Military capabilities needed from sensors are day and night all-weather capability, long-range operation, rapid and large area search capability, detection of low signature